

# Lab 1: LLMs and RAG

## Handling of ASR\_NOINPUT

The system kept entering an infinite loop when there was no response. Hence, I initialized a counter variable to 0. Whenever there is no response from the user, the counter gets incremented by 1. After 3 consecutive no inputs, it moves to the Done state. The counter is reset back to 0 every time the user speaks.

## Experiments with RAG

### One shot vs Multi turn

The RAG Model worked well for one shot queries and also multi turn queries. During testing, the system correctly identified that "their" in the second query refers to "Feelgood" of the previous question. This means that it correctly identifies the context and can continue conversations to some extent. This is because the entire conversation history is stored in messages which is an Array containing the system, user and assistant messages.

U : Can you tell me what is feel good? 0.81082356

S : According to the text, Feelgood is a student healthcare service that can be contacted by phone, email, or through their app, and it offers services such as medical appointments and cancellations.

U : What is their phone number? 0.8311845

S : The phone number for Feelgood is 031-786 43 22, and it's available on weekdays from 8 am to 5 pm.

### Number of retrieved results

I experimented my system with setting limit: 1 and limit: 5. I noticed that when retrieving the top 1 result, direct questions worked well. For eg: Fellgood's phone number retrieved the correct answer. However it could provide insufficient information for broader question's like below.

U : OK, uh, where is the university library located? 0.7586915

S : The University Library's location can be found through the University of Gothenburg's website.

U : MMM, can you give me the locations? 0.7676247

S : University Library locations include: Main Library, Medicinal History Library, and Language & Literature University Library, all on the University of Gothenburg's campus.

It was quite interesting to see that the retrieved top 1 result did not contain the library locations. Rather it was **ragResult: 'This is frequently found in the Contact, Help or About pages.', counter: 0}** But the LLM still generated specific library names. This could be because LLM knew from the conversation history that the locations referred to the University Library

## Filtering and Re-ranking

In our first seminar, we discussed the paper **Exploring Retrieval Augmented Generation Approaches for Natural Language Question Understanding** which introduces a filtering state after retrieval. LLM filters out the information that is not relevant from the top k retrieved results. In my current implementation, i am retrieving 5 results and passing all the 5 to the LLM. This worked pretty well for my dataset. However i think this approach is not practically possible for a much larger dataset or a real world scenario, since retrieving more results may increase the amount of context to be passed into LLM. Hence a filtering stage could be a better option.

Qdrant doc says that re-ranking can be used after the retrieval to improve the results. From my experiment with limit: 1, I noticed that the highest ranked result was not the most useful one. The top retrieved result “This is frequently found in the Contact, Help or About pages” did not contain the information needed to answer my question. Hence, i think re-ranking could be useful in such cases.